

## Some Descriptive Set Theory

### Polish Spaces and Standard Borel Spaces

**Definition:** A topological space  $X$  is called a *Polish space* if it is separable and completely metrizable.

**Theorem:** Let  $X$  be an uncountable Polish space. Then,  $X$  has a closed subset  $Y$  which is a Cantor set.

*Proof.* Let  $d$  be a compatible metric on  $X$ . Let  $Z$  be the subset of  $X$  consisting of all points  $x$  such that each neighborhood of  $x$  is uncountable. Then,  $Z$  is closed.

For each  $x \in X \setminus Z$ , let  $r_x$  denote the supremum of all  $r \in (0, 1)$  such that the open ball centered at  $x$  with radius  $r$  is countable.

Let  $S$  be a countable dense subset of  $X \setminus Z$ . Then, the open balls around  $x \in S$  with radius  $\frac{1}{2}r_x$  for each  $x \in S$  cover  $X \setminus Z$ . In particular, this means that  $X \setminus Z$  is countable. Therefore, we may assume that  $Z = X$ .

By a recursive procedure, we may construct pairwise disjoint closed balls  $B_{n,1}, B_{n,2}, \dots, B_{n,2^n}$  such that each  $B_{n,k}$  has a positive radius no larger than  $1/n$  and contains two of the sets  $B_{n+1,1}, B_{n+1,2}, \dots, B_{n+1,2^{n+1}}$ . Then, we have that the set

$$C = \bigcap_{n=1}^{\infty} \bigcup_{k=1}^{2^n} B_{n,k}$$

is a closed subset of  $X$  that is a Cantor set. □

**Definition:** A measurable space  $(X, \mathcal{S})$  is called a standard Borel space if it is isomorphic as measurable spaces to  $(Y, \mathbf{B}(Y))$ , where  $\mathbf{B}(Y)$  denotes the Borel  $\sigma$ -algebra, for some Polish space  $Y$ . Equivalently, there is a Polish topology  $\mathcal{T}$  on  $X$  such that  $\mathcal{S} = \mathbf{B}(\mathcal{T})$ .

## Countable Borel Relations, Invariant Measures, and Graphings

### Equivalence Relation Preliminaries

First, we collect some definitions.

**Definition:**

- A relation  $R$  on a set  $X$  is a set of ordered pairs  $R \subseteq X^2$ . We will write  $xRy$  to denote  $(x, y) \in R$ . We also define the *sections*

$$R_x = \{y : (x, y) \in R\}$$

$$R^y = \{x : (x, y) \in R\}$$

and the inverse relation

$$R^{-1} = \{(y, x) : (x, y) \in R\}.$$

- If  $X$  is equipped with a Borel structure, then the relation  $R$  (i.e., equivalence relations, partial orders, total orders, etc.) is called Borel if  $R \subseteq X^2$  is Borel.
- We will let  $\text{dom}(f)$ ,  $\text{ran}(f)$ , and  $\text{graph}(f)$  denote the domain, range, and graph of  $f$  respectively.
- A graph  $G$  with vertex set  $X$  is a non-reflexive, symmetric relation on  $X$ . The neighbors of  $x \in X$  in the graph  $G$  are the set of all  $y \in X$  such that  $(x, y) \in G$ . The cardinality of the set of neighbors of  $x$  is called the degree of  $x$ , denoted  $d_G(x)$ .
- A path in  $G$  is a finite sequence of vertices  $(x_0, x_1, \dots, x_n)$  where  $(x_i, x_{i+1}) \in G$ , and  $x_i \neq x_j$  whenever  $i \neq j$  except for possibly  $x_n = x_0$ . The equivalence classes of the relation  $xEy$  if and only if there exists a path from  $x$  to  $y$  defines the connected components of the graph. If a graph has one connected component, we say the graph is connected.

- If  $x_n = x_0$  with  $n \geq 3$ , then we call the path a cycle. A graph that has no cycles is called acyclic, and a connected acyclic graph is called a tree. A rooted tree is a tree with a distinguished vertex  $x_0$  called its root.
- If  $X$  is a standard Borel space and  $E$  is a Borel equivalence relation on  $X$ , then we say the relation is countable if  $[x]_E$  is countable for each  $x \in X$ .
- If  $\Gamma \curvearrowright X$  is a countable group acting on  $X$  such that  $\Gamma$  preserves the Borel structure of  $X$ , then we define the orbit equivalence relation to be  $x E_\Gamma^X y$  if and only if there exists  $g$  such that  $g \cdot x = y$ .
- The orbit space  $X/\Gamma$  is defined to be the quotient  $X/E_\Gamma^X$ .
- If  $E$  is a countable Borel relation, then the *full group* of  $E$ , denoted  $[E]$ , is the set of Borel automorphisms whose graphs are contained in  $E$ . The set of partial Borel automorphisms is denoted  $[[E]]$ ; that is,

$$[[E]] = \{\phi: A \rightarrow B : \phi \text{ is a bijection, } A, B \subseteq X \text{ are Borel, } \text{graph}(\phi) \subseteq E\}.$$

- We call a countable equivalence relation  $E$  on  $X$  *aperiodic* if every equivalence class  $[x]_E$  is infinite.
- Given Borel equivalence relations  $E$  and  $F$  on  $X$  and  $Y$  respectively, we say  $E$  is *Borel reducible* to  $F$ , denoted  $E \leq_B F$ , if there is a Borel map  $f: X \rightarrow Y$  such that  $x E y$  if and only if  $f(x) F f(y)$ .

**Example:** If  $X^\Gamma$  denote the space of functions of  $\Gamma$  onto  $X$ , then the left shift action of  $\Gamma$  on  $X^\Gamma$  is defined by  $\gamma \cdot p(\delta) = p(\gamma^{-1}\delta)$ .

If  $G$  is a standard Borel group and  $\Gamma \subseteq G$  is a countable discrete subgroup (also known as a lattice), then the action of  $\Gamma$  onto  $G$  by left translation, taking  $\gamma \cdot g = \gamma g$ , yields the quotient space  $G/\Gamma$  given by the set of *right* cosets of  $\Gamma$ .

**Example:** The following are examples of countable Borel equivalence relations.

- If  $X = 2^\mathbb{N}$ , then the *eventual equality* relation says that  $x E_0 y$  if and only if there exists  $m$  such that for all  $n \geq m$ , we have  $x_n = y_n$ . The *tail equivalence* relation says that  $x E_t y$  if and only if there exist  $m$  and  $k$  such that for all  $n$ , we have  $x_{m+n} = y_{k+n}$ .
- If  $X = \mathbb{R}$ , then the *Vitali relation* says that  $x E_v y$  if and only if  $x - y \in \mathbb{Q}$ .
- If  $X = \mathbb{R}_+$ , then the *commensurability relation* says that  $x E_c y$  if and only if  $x/y \in \mathbb{Q}$ .
- If  $k \geq 2$  and  $X$  is the space of subshifts of  $k^\mathbb{Z}$ , where a subshift is a nonempty closed subset of  $k^\mathbb{Z}$  invariant under the shift map  $(S(x))_i = x_{i-1}$ .

This is a compact subspace of the hyperspace of compact subsets of  $k^\mathbb{Z}$ , so it is a compact and metrizable space. Define two subshifts to be isomorphic if there is a homeomorphism between the two closed subsets that commutes with the shift, and define an equivalence relation as such.

**Theorem (Feldman–Moore):** Let  $E$  be a countable Borel equivalence relation on  $X$ . Then, there is a countable group  $\Gamma$  and a Borel action of  $\Gamma$  on  $X$  such that  $E_\Gamma^X = E$ .

**Proposition (Marker Lemma):** Let  $E$  be an aperiodic countable Borel equivalence relation on  $X$ . Then, there is a sequence  $\{S_n\}_{n \geq 1}$  of Borel sets  $S_n \subseteq X$  such that

- $S_0 \supseteq S_1 \supseteq S_2 \supseteq \dots$ ;
- $\bigcap_{n \geq 1} S_n = \emptyset$ ;
- every  $S_n$  is a complete section for  $E$  (i.e., it has nonempty intersection with each equivalence class of  $E$ ).

Such a sequence is called a *vanishing sequence of markers*.

*Proof.* Via some results from descriptive set theory, we may assume that the equivalence relation  $E$  lives on the set  $X = 2^\mathbb{N}$ . Then, we let  $s_n(x)$  be the lexicographically smallest  $s \in 2^\mathbb{N}$  such that  $|[x]_E \cap \mathcal{N}_s| = \infty$ , where  $\mathcal{N}_s$  denotes the cylinder set in  $X$  corresponding to  $s$ .

Define  $x \in A_n$  if and only if  $x|_n = s_n(x)$ . Then, we have that  $\{A_n\}_{n \geq 1}$  is a decreasing sequence of Borel sets which intersects each class of  $E$  at infinitely many points. Then,  $A = \bigcap_{n \in \mathbb{N}} A_n$  intersects each class of  $E$  in at most one point, so that  $B_n = A_n \setminus A$  is the vanishing sequence of markers as desired.  $\square$

## Invariant Measures

**Definition:** If  $E$  is a countable Borel equivalence relation on  $X$ , then we say  $\mu$  is  $E$ -invariant if for all  $f \in [E]$ , we have  $f_*\mu = \mu$ , where  $f_*\mu$  is the pushforward measure, defined by

$$f_*\mu(A) = \mu(f^{-1}(A)).$$

**Proposition:** The following are equivalent:

- (a)  $\mu$  is  $E$ -invariant;
- (b)  $\mu$  is  $\Gamma$ -invariant for any countable group  $\Gamma$  such that  $E = E_\Gamma^X$ ;
- (c)  $\mu$  is  $\Gamma$ -invariant for some  $\Gamma$  such that  $E = E_\Gamma^X$ ;
- (d) for all  $\phi \in [[E]]$ , we have  $\mu(\text{dom}(\phi)) = \mu(\text{ran}(\phi))$ .

If  $\mu$  is invariant under  $E$ , then we may define a measure  $\bar{\mu}$  on  $E$  by taking

$$\bar{\mu}(A) = \int |A_x| d\mu(x),$$

where again,

$$A_x = \{y : (x, y) \in A\}.$$

It is also possible to define

$$\bar{\mu}'(A) = \int |A^y| d\mu(y),$$

but since  $\mu$  is invariant,  $\bar{\mu} = \bar{\mu}'$ . This follows from the fact that

$$E = \bigcup_{i \in \mathbb{N}} \text{graph}(g_i)$$

for some collection of relation-preserving automorphisms  $g_i \in [E]$ . If  $A \subseteq E$  is any Borel subset, then

$$A = \bigcup_{i \in \mathbb{N}} (\text{graph}(g_i) \cap A),$$

so there are  $f_i \in [[E]]$  such that  $\text{graph}(f_i) = \text{graph}(g_i) \cap A$ . Therefore, we may write any  $A \subseteq E$  as a countable disjoint union of graphs of functions  $f \in [[E]]$ . Therefore, it suffices to show that  $\bar{\mu}(\text{graph}(f)) = \bar{\mu}'(\text{graph}(f))$  for any  $f \in [[E]]$ . If  $C = \text{dom}(f)$  and  $D = \text{ran}(f)$ , then  $\bar{\mu}(\text{graph}(f)) = \mu(C)$  and  $\bar{\mu}'(\text{graph}(f)) = \mu(D)$ , so by the definition of  $[[E]]$ , we are done.

Now, we may define an equivalence relation  $\tilde{E}$  on  $E$  by taking  $(x, y)E(z, w)$  if and only if  $xEz$ , or equivalently,  $xEyEzEw$ .

**Proposition:** The invariant measure  $M$  on  $E$  is  $\tilde{E}$ -invariant.

*Proof.* Let  $\Gamma$  act on  $X$  such that  $E = E_\Gamma^X$ . Then, if we define the action  $(g, h) \cdot (x, y) = (g \cdot x, h \cdot y)$ , then we have that  $E_{\Gamma^2}^E = \tilde{E}$ , so it suffices to show that the action preserves  $M$ .

For this, if  $A \subseteq E$  is Borel, we have

$$M((g, h) \cdot A) = \int |\{(g \cdot x, h \cdot y) : (x, y) \in A\}| d\mu(x).$$

Setting  $B = (g, h) \cdot A$ , then we have  $y \in B_x$  if and only if  $(x, y) \in B$ , which holds if and only if  $(g^{-1} \cdot x, h^{-1} \cdot y) \in A$ , which holds if and only if  $h^{-1} \cdot y \in A_{g^{-1} \cdot x}$ . Therefore,  $B_x = h \cdot A_{g^{-1} \cdot x}$ , so that

$$\begin{aligned} M((g, h) \cdot A) &= \int |A_{g^{-1} \cdot x}| d\mu(x) \\ &= \int |A_x| d\mu(x) \\ &= M(A), \end{aligned}$$

where we use the fact that  $\mu$  is invariant under  $\Gamma$ . □

## Graphings and Cost

**Definition:** If  $E \subseteq X^2$  is a countable Borel equivalence relation, then a (Borel) *graphing* of  $E$  is a graph  $G$  such that the connected components of  $G$  are the equivalence classes of  $E$ . We say that  $G$  generates  $E$ .

**Definition:** A Levitt graph (L-graph) is a countable family  $[[E]] \supseteq \Phi = \{\varphi_i\}_{i \in I}$  of partial Borel isomorphisms  $\varphi_i: A_i \rightarrow B_i$ , where  $A_i, B_i$  are Borel subsets of  $X$ .

We call  $\Phi$  *finite* if  $I$  is finite. We say  $\Phi$  is an Levitt graphing (L-graphing) if  $\Phi$  generates  $E$  in the sense that  $xEy$  if and only if  $x = y$  or there is a finite set  $i_1, \dots, i_k \in I$  and  $\varepsilon_1, \dots, \varepsilon_k \in \{1, -1\}$  such that  $x = \varphi_{i_1}^{\varepsilon_1} \cdots \varphi_{i_k}^{\varepsilon_k}(y)$ .

For every L-graph  $\Phi$ , there is an associated graph  $G_\Phi$  that generates the same equivalence relation. We say  $xG_\Phi y$  if and only if  $x \neq y$  and there exists  $i$  such that  $\varphi_i(x) = y$  or  $\varphi_i(y) = x$ .

Conversely, for every graph  $G$ , there is an L-graph  $\Phi_G$  such that  $G = G_{\Phi_G}$ . This follows from fixing a Borel (total) ordering  $<$  on  $X$ , and finding a family of partial Borel functions  $\{f_i\}_{i \in \mathbb{N}}$  with disjoint graphs such that for all  $x \in X$ , we have

$$\{(x, y) : xGy \text{ and } x < y\} = \bigcup_{i \in \mathbb{N}} \text{graph}(f_i).$$

Since the  $f_i$  are countable-to-one, we have  $\{g_{i,j}\}_{j \in K_i}$  a family of partial Borel automorphisms with disjoint graphs such that

$$\text{graph}(f_i)^{-1} = \bigcup_{j \in K_i} \text{graph}(g_{i,j})$$

(Recall that we define the inverse of a relation to be the coordinates flipped.) Define  $\Phi_G = \{g_{i,j}\}_{i,j}$ .

Considering the case of an  $E$ -invariant  $\mu$ , if  $G$  is a graphing of  $E|_A$  for some conull  $E$ -invariant Borel subset  $A$ , then we say  $G$  is a graphing of  $E$  almost everywhere. Similarly, we may define an L-graphing almost everywhere.

Now, we will discuss the theory of cost.

**Definition:** Let  $E$  be a countable Borel equivalence relation on  $X$ , and let  $\mu$  be an  $E$ -invariant measure with corresponding measure on  $E$  given by  $M$ . If  $G \subseteq E$  is a subset, then we define its *cost* to be

$$C_\mu(G) = \frac{1}{2}M(G).$$

Observe that  $0 \leq C_\mu(G) \leq \infty$ , and  $0 < C_\mu(G)$  if  $G$  is a graphing of  $E$  provided that it's not the case that almost every  $E$ -class is a singleton. This follows from the fact that  $G$  is a complete section for the relation  $\widetilde{E}$  on  $E$ , meaning that  $M(G) > 0$ .

Now, if  $\Phi = \{\varphi_i\}_{i \in I}$  is an L-graph, define its cost by taking

$$C_\mu(\Phi) = \sum_{i \in I} \mu(\text{dom}(\varphi_i)) = \sum_{i \in I} \mu(\text{ran}(\varphi_i)).$$

Then, we have

$$C_\mu(\Phi) = \frac{1}{2} \int \sum_{i \in I} (\chi_{A_i}(x) + \chi_{B_i}(x)) d\mu(x),$$

where we have  $A_i = \text{dom}(\varphi_i)$  and  $B_i = \text{ran}(\varphi_i)$ .

## References

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